

Question 1 - Data Pre-processing and Exploration

Overall Data Analysis

In the pre-processing stage, an additional indicator, 'Gross_capital_formation', was added to the initial dataset of 10 indicators. The data was downloaded from the World Bank API using the provided Python script.

The data exploration has shown the amount of missing values for each indicator presented in Table 1. The highest percentage of missing values was observed for the literacy rate and poverty ratio, with 88% and 86% of missing values, respectively.

Table 1: Missing Values Summary

Column Name	Missing Values	Percentage of Missing Values	Data Type
Country	0	0%	str
Date	0	0%	str
GDP	3001	30.7%	float64
Population	10	0.1%	float64
Life Expectancy	10	0.1%	float64
Literacy Rate	8581	87.8%	float64
Unemployment Rate	3436	35.2%	float64
Fertility Rate	10	0.1%	float64
Poverty Ratio	8436	86.4%	float64
Primary School Enrolment Rate	3455	35.4%	float64
Gross Capital Formation	2876	29.4%	float64
Energy Use	4443	45.5%	float64
Exports	3730	38.2%	float64

The dataset contains **107,448** numerical observations from **11** different econometric variables for **222** different countries. The percentage of missing values for the whole dataset is **35.4%**, which is a relatively high value, indicating that a careful imputation process is required.

Analysis for United States, Russian Federation, China and Brazil

Table 2: Missing values summary for numeric columns by country

Country	Number of Missing Values	Percentage of Missing Values
United States	136	28.1%
Brazil	131	27.1%
China	150	31.0%
Russian Federation	116	24.0%

Data Imputation

The exploratory data analysis revealed that the dataset contained a very high proportion of missing values, reaching up to 88% for some variables. For this reason, simple imputation methods such as backward and forward fills were not considered appropriate. To address this issue, the following framework was applied. First, the data were separated by country and by feature so that each series could be processed individually. Next, the maximum number of consecutive missing years was calculated for each case. For the maximum gap of 1 or between 6 and 15 years, linear interpolation was used as a smooth and reasonable estimator of the missing values. For gaps of 2, 3, 4, and 5 years, cubic interpolation was applied instead, as it can better capture curved trends over short gaps than a straight-line estimate. Finally, for features with more than 15 consecutive missing values, the kNN imputer was used and fitted on the global dataset, since the lack of sufficient observations within the feature itself did not allow interpolation-based methods to be applied reliably. To avoid data leakage from the kNN imputer, data was split into train, validation, and test split prior to imputation and saved in different files for future modelling.

Further data preprocessing was creating sequences of 5-year windows that overlap by 4 years, ending up with 44 sequences each size of 55.

Question 2 - MLP Implementation

The log transformation of GDP across all the countries was used to assign each country one of the following labels: Under-developed, Developing, Emerging, Developed. This process was done using quantile binning, meaning that each class ended up with roughly 25% of the data, ensuring class balance. Quantile binning borders were calculated on the training data and later used on testing data. This, together with GDP removal ensured no data leakage and unrealistic results.

The MLP implementation was done in three consecutive steps:

First, hyperparameter tuning was performed through GridSearch to find the optimal amount of hidden layers and approximately identify the best model parameters. It was run across different combinations of parameters listed below:

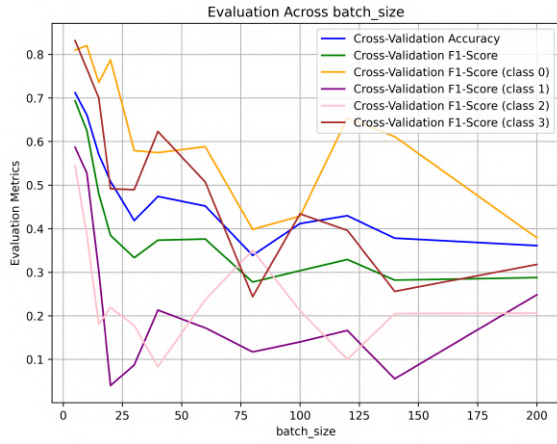
- hidden layer sizes:[(32), (64), (64,32), (128,64), (128,64,32), (64, 32, 16)]
- learning rate:[0.0001, 0.005, 0.01, 0.05, 0.1]
- batch size:[5, 10, 15, 20, 50, 100]
- number of epochs:[20, 40, 60, 80, 100, 200, 300, 400, 600]

Table 3: Top 10 MLP Configurations Through-Out GridSearch Based on Accuracy Metric

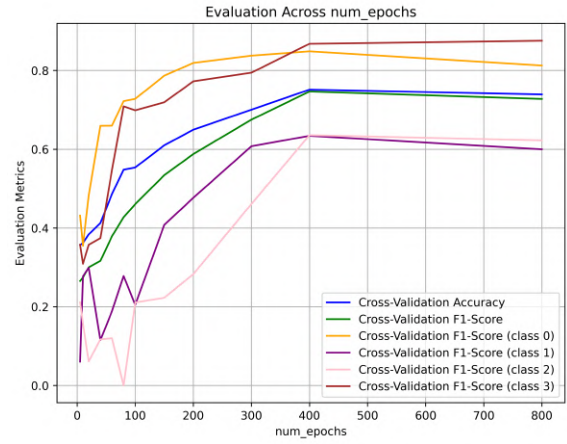
Layer Sizes	Learning Rate	Batch Size	Num Epochs	CV Accuracy	Macro F1	F1 (Class 0)	F1 (Class 1)	F1 (Class 2)	F1 (Class 3)
(50, 64, 32, 16, 4)	0.100	100	300	0.779841	0.775858	0.857807	0.670065	0.669841	0.905719
(50, 64, 32, 16, 4)	0.001	5	400	0.774127	0.772390	0.829735	0.690737	0.699049	0.870038
(50, 64, 32, 4)	0.050	50	300	0.773810	0.767953	0.842513	0.702170	0.652989	0.874140
(50, 64, 4)	0.050	15	80	0.773810	0.768052	0.845870	0.671096	0.670643	0.884598
(50, 64, 32, 16, 4)	0.010	20	600	0.768571	0.765358	0.850378	0.695065	0.649735	0.866254
(50, 128, 64, 4)	0.050	15	60	0.768571	0.762236	0.842895	0.647464	0.662222	0.896362
(50, 64, 32, 16, 4)	0.005	10	600	0.768413	0.765141	0.859180	0.709538	0.642705	0.849140
(50, 128, 64, 32, 4)	0.050	5	20	0.768254	0.761218	0.831111	0.666754	0.663978	0.883029
(50, 64, 32, 4)	0.010	15	300	0.768095	0.762495	0.847601	0.679817	0.648421	0.874140
(50, 128, 64, 4)	0.010	10	100	0.768095	0.765499	0.820905	0.664957	0.678097	0.898039

The GridSearch has identified the optimal model parameters presented in Table 3, resulting in best performance for the model that uses 3 hidden layers with 64, 32 and 16 neurons respectively.

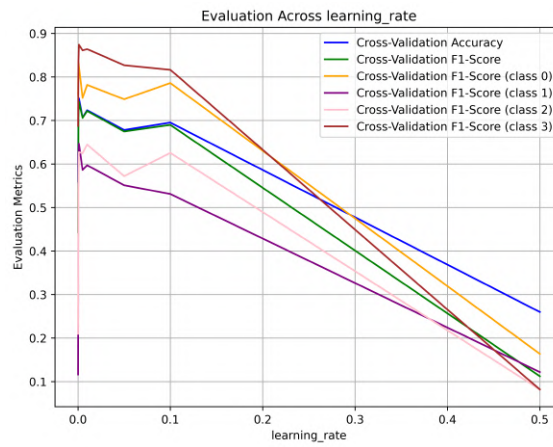
The second step of the model implementation involved performing a 5-fold cross-validation on the training set to further analyse the hyperparameters on (64, 32, 16) hidden layers. For each configuration, cross-validation, accuracy, F_1 -score of each class, and an aggregated F_1 -score across all 4 classes were recorded, as shown in Figure 1, to identify the optimal batch size, number of epochs and learning rate.



(a) Evaluation Across Batch Size



(b) Evaluation Across Number of Epochs



(c) Evaluation Across Learning Rate

Figure 1: Evaluation Across Hyperparameters

The plots in Figure 1 show the effect of batch size, learning rate and the number of epochs, which help identify the best parameters for the final model selection. The process has further selected a model with batch size 5, 400 epochs and learning rate 0.001. The selected model is extremely difficult with large number of training step due to the small batch size. The cross-validation results indicate that the model performs best when trained with a small batch size, a relatively small step of weight update and a higher number of epochs. Aggregated F_1 -score shows very close results to the accuracy metrics across model selection process, but there is a clear trade-off in predictions of class 0 and 1 indicating slight presence on class imbalance.

The final model configuration was chosen by considering the results of the hyperparameter tuning using GridSearch and the training vs. testing dynamics across not only accuracy but also the F_1 -score for each of the classes. Both models, recommended by the GridSearch and cross-validation were further fitted on the testing data to understand the generalisation of the model. The results presented in Figure 2 have shown that the best GridSearch model with batch size 5, number of epochs 300 and a learning rate of 0.1, have shown almost 100% during training but fail to generalise resulting in approximately 65% accuracy on a testing subset.

The 2 best GridSearch models, which were also proposed by a cross-validation procedure (see

Figure 3), has shown good generalisation and good overall performance, with training and testing both having approximately 80% accuracy after the training is done. This observation further confirms that a large number of small steps, tends to improve the performance of this MLP model.

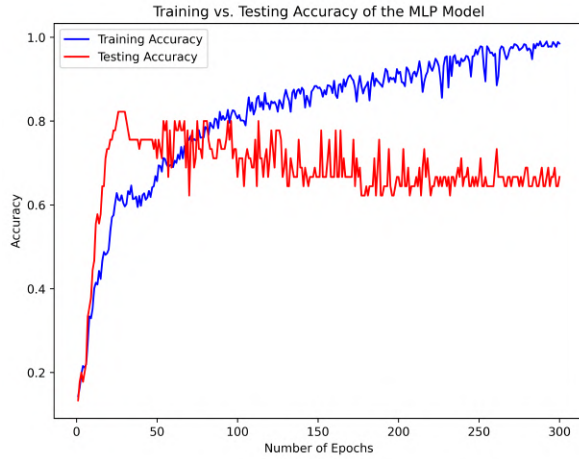


Figure 2: Training vs. Testing Accuracy Plot
best model by grid search

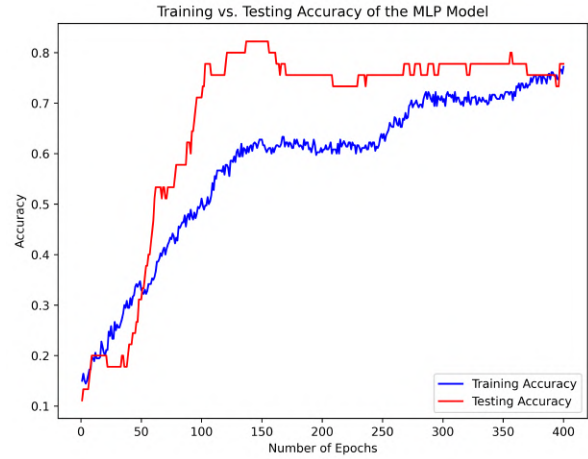


Figure 3: Training vs. Testing Accuracy Plot
best model after cross-validation

Table 4: Optimal MLP Parameters

Parameter	Value
Batch Size	5
Learning Rate	0.001
Number of Epochs	400

After the selection of the model parameters, the final model was trained with results shown in Table 5. The model has achieved an overall accuracy of 78% and an F_1 -score of 76%, suggesting a strong classification performance for all 4 classes. Based on the table, class 3 appears to have the best performance with a perfect score in precision and specificity and a high F_1 -score of 92.9%, indicating that the feature space of the class is easily identifiable by the model. Classes 0 and 2 show reasonably good performance, whereas class 1 shows poor performance with low precision of 50% and an F_1 -score of 59%, suggesting that the model classifies many false positive cases, and therefore struggles to distinguish between the classes, most likely as a result of overlapping feature distributions between neighbouring GDP-based classes.

Table 5: MLP Country Classification Performance Metrics

Class	Precision	Recall	Specificity	F1-Score
0	0.8182	0.8182	0.9412	0.8182
1	0.5000	0.7143	0.8684	0.5882
2	0.7273	0.6667	0.9091	0.6957
3	1.0000	0.8667	1.0000	0.9286
Overall Accuracy: 78%				Overall F1 Score: 76%

Question 3 - Time-Series Forecast

This section compares the time series forecasting performance of three deep learning models. The last 15 years of data were held out, with 10 years used for validation and the remaining 5 years used for testing. Sequential data were then constructed using 10-year historical windows as input and 5-year periods as the prediction horizon for each model. A log transformation was applied to GDP, and the data was standardised separately for each country. Model selection for each framework involved an initial grid search over a set of hyperparameters, with constant batch size = 36 and learning rate = 0.01, followed by cross-validation of the best-performing configurations to further tune the learning rate, batch size, and number of epochs. The Mean Square Error was used as metric to have more penalty for larger errors. The model selection process may not cover all possible combinations of parameter due to the computational limitations, but can be considered as a good approximation.

LSTM

The grid search for LSTM model was performed over the following hyperparameters space, resulting in 108 iterations.

- hidden size: [24, 48, 64, 80, 128, 256],
- number of layers: [1, 2, 4],
- drop out: [0.1, 0.3],
- number of epochs: [5, 10, 20]

The performed grid search revealed that across epochs of 5, 10 and 20, top models use hidden size of 256 and 2 LSTM models stacked on each other.

Table 6: Top 5 GridSearch Results of the LSTM Model

Batch Size	Hidden Size	Num. of Layers	Drop Out	Learning Rate	Num. of Epochs	Train Score	Validation Score
36	256	1	0.1	0.01	5	0.1172	0.6255
36	64	4	0.3	0.01	5	0.1675	0.6316
36	48	1	0.3	0.01	5	0.1124	0.6330
36	128	2	0.3	0.01	5	0.1324	0.6462
36	128	1	0.1	0.01	5	0.1092	0.6614

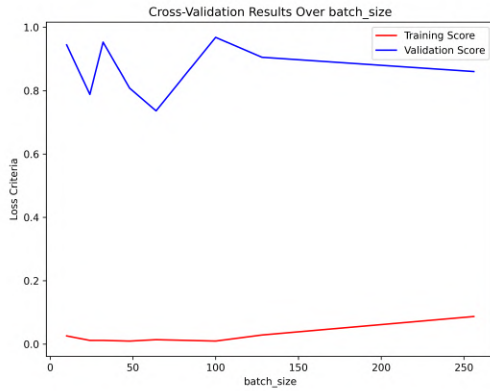
The plots in Figure 4 display cross-validation results that been applied on model selected by Grid search and have identified the best hyperparameters shown below:

- batch size = 64,
- number of epochs = 200
- learning rate = 0.005

Based on Figure 4a, the changes in batch size from 5 to 250 did not lead to major differences in performance, but instead produced moderate fluctuations in the MSE values, ranging from approximately 0.8 to 1.0. Among the tested values, a batch size of 64 achieved the best validation performance and was therefore selected. The learning rate showed clear superior performance at 0.005, which was chosen as the optimal value. As shown in Figure 4c, 200 epochs appeared to provide the best balance between learning the underlying patterns in the data and avoiding overfitting to the training set. The selection process resulted in the following parameters of

LSTM model:

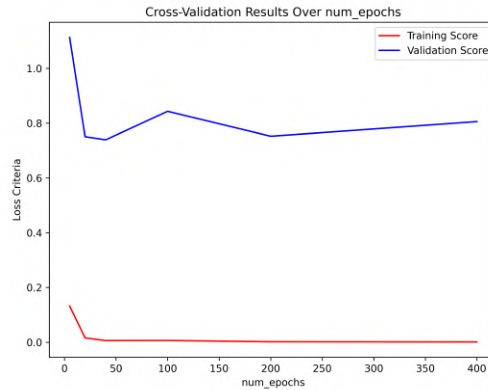
- hidden size = 256
- number of layers = 2
- drop out = 0.1
- number of epoches = 200
- learning rate = 0.005
- batch size = 64



(a) CV MSE Scores Across Batch Size



(b) CV MSE Scores Across Learning Rate



(c) CV MSE Scores Across A Number of Epochs

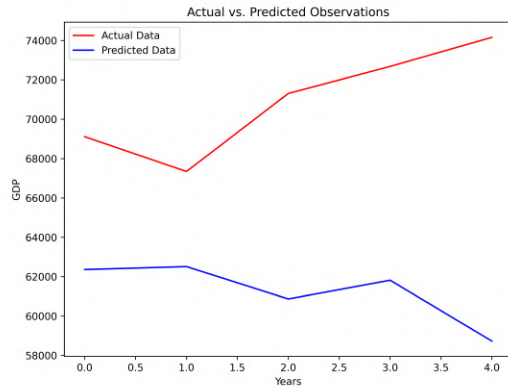
Figure 4: LSTM CV MSE Scores Across Hyperparameters

The combination of results suggest that model is actually learning from the data, however based on the model selection process and the forecasting results obtained, its ability to generalise is limited, and it fails to capture the GDP trend in most cases.

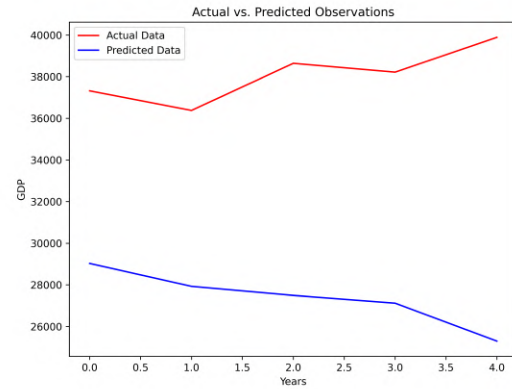
The most accurate forecasts were obtained for Brazil and the USA, with forecast errors of 13.2% and 13.5%, respectively. As shown in Figures 5c and 5a, the model appears to capture some of the short-term fluctuations in the data, but it still fails to follow the overall trend. In contrast, the model performed very poorly for China, producing a forecast error of 73.9%, which suggests that this framework is not suitable for that country. For the Russian Federation, the forecasts were consistently underestimated, with a MAPE value of 28.0%.

Table 7: LSTM Model Loss Criteria Results

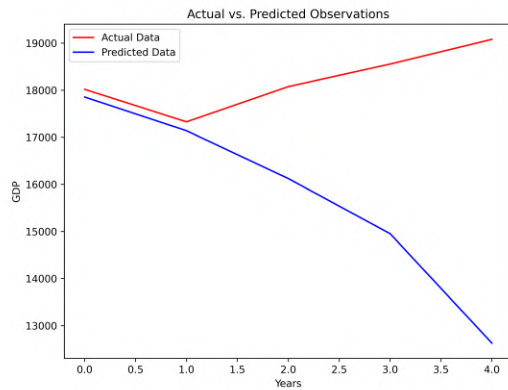
Country	MAE	MAPE	MSE
United States	9664.3	13.5%	106835587.9
Russian Federation	10712.2	28.0%	120021113.2
China	15598.1	75.5%	246108418.1
Brazil	2471.4	13.2%	11692866.6
Overall testing dataset results:	1.05		2.2665



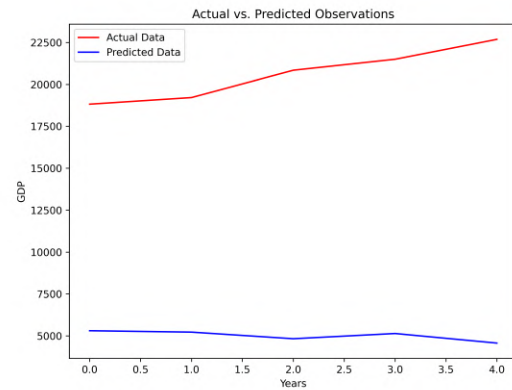
(a) United States - Actual vs. Predicted LSTM Results



(b) Russian Federation - Actual vs. Predicted LSTM Results



(c) Brazil - Actual vs. Predicted LSTM Results



(d) China - Actual vs. Predicted LSTM Results

Figure 5: Actual vs. Predicted LSTM Results

Table 8: Actual vs Predicted Values by Country for LSTM

Country	Type	Year 1	Year 2	Year 3	Year 4	Year 5
United States	Predicted	62360.8	62511.3	60865.0	61817.3	58724.1
	Actual	69112.6	67342.1	71307.4	72679.3	74158.7
Russian Federation	Predicted	29030.4	27928.6	27494.8	27122.2	25299.1
	Actual	37319.3	36376.3	38638.4	38214.5	39887.3
China	Predicted	5308.3	5224.8	4830.5	5141.8	4569.3
	Actual	18820.5	19215.2	20843.3	21499.4	22687.1
Brazil	Predicted	17854.9	17139.2	16123.2	14952.8	12628.4
	Actual	18018.6	17327.5	18075.7	18554.0	19079.8

CNN-LSTM

The model selection for CNN LSTM was initiated with Grid Search over the following set of hyper parameters, resulting in 540 iterations.

- hidden size: [24, 48, 64, 80, 128, 256],
- number of layers: [1, 2, 4],
- drop out: [0.1, 0.3],
- number of epochs: [5, 10, 20],
- cnn layers: multiple CNN specifications with 1 or 2 layers

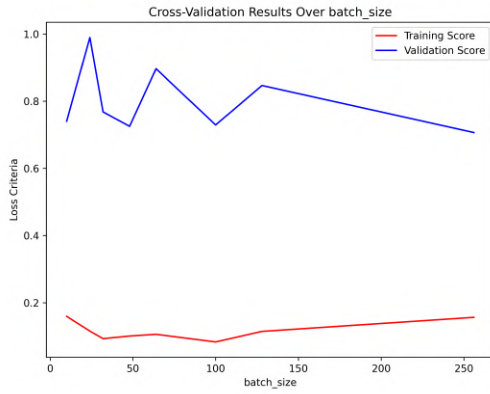
The results identified that CNN with 2 layers, 64 and 32 filters, respectively, 24 hidden size and 0.3 drop out have provided best results on the validation set. But this model shows comparatively bad loss on the training dataset, therefore the next few models were considered as well as tested. These models tend to deteriorate with increasing number of epochs so the top one model by evaluation metrics was further used for cross-validation.

Table 9: Top 5 GridSearch Results of the CNN-LSTM Model

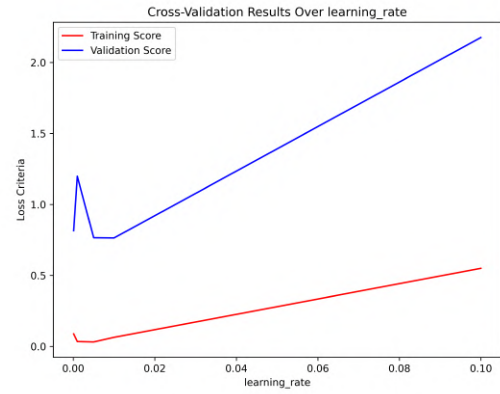
Hidden Size	Num. of Layers	Drop Out	Num. of Epochs	Train Score	Validation Score
24	1	0.3	5	0.1835	0.6041
128	1	0.1	10	0.1174	0.6061
48	1	0.1	20	0.0884	0.6281
256	1	0.1	20	0.1140	0.6283
48	1	0.3	10	0.1396	0.6408

Cross-validation process was in the following order, with consecutive hyperparameter update. Batch size (48 - best) -> hidden size (128 - best)-> number of epochs (100)-> learning rate(0.05) Based on 6a, the best validation performance occurs when the batch size is at moderate values ,especially at batch size of 48, making it the optimal size. Figure 6b displays an increase in the validation loss as the learning rate increases, suggesting that smaller learning rates provide more stability and better generalisation. In particular, a learning rate of 0.005 was selected. Similarly, the optimal number of epochs, shown in Figure 6c, appears at 100 epochs where the MSE validation loss is at its smallest value. Lastly, the optimal hidden size based on Figure 6d

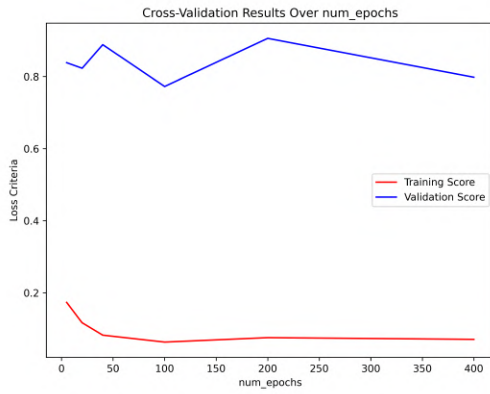
is 128, where the MSE loss is slightly above 0.6.



(a) CV MSE Scores Across Batch Size



(b) CV MSE Scores Across Learning Rate



(c) CV MSE Scores Across A Number of Epochs



(d) CV MSE Scores Across Hidden Size

Figure 6: CNN-LSTM CV MSE Scores Across Hyperparameters

Based on the Grid Search and cross-validation, the best model configuration appears to have the following parameters:

- hidden size = 128
- number of layers = 1
- drop out = 0.3
- learning rate = 0.005
- number of epochs = 100
- batch size = 48
- number of CNN layers = shown in Figure 10

Table 10: CNN Layers

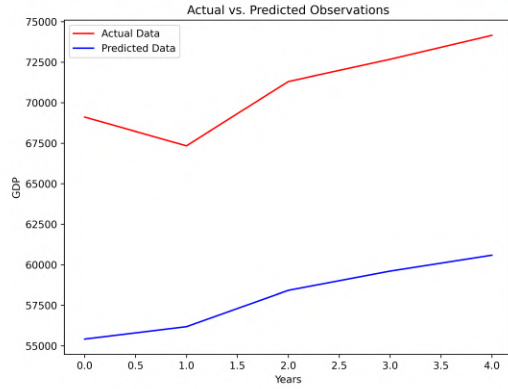
Layer	Num. of Filters	Kernel Size	Padding Size	Stride Size	Pool Type	Pool Size
1	64	3	1	1	Max	3
2	32	3	1	1	Adaptive	10

Overall, the CNN-LSTM model led to a deterioration in forecasting performance across almost all countries. The MAE calculated over the full testing dataset increased from 1.05 to 1.07, suggesting that the feature extraction introduced by the CNN component reduced the LSTM's ability to capture temporal relationships in the data. However, visual inspection of the forecasts for Brazil indicates that the CNN-LSTM model provides a better representation of both the overall trend and the fluctuations for this country. This is supported by the reduction in MAPE from 13.2% to 11.1%. In contrast, all other countries experienced a deterioration in performance, with forecast errors increasing by approximately 5% to 10%. For the USA, the predicted trend appears to improve visually, although the overall GDP level remains underestimated.

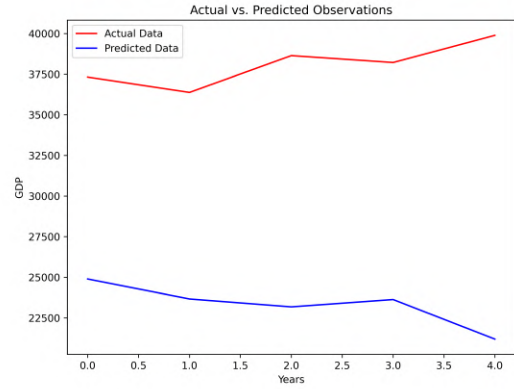
Overall, the CNN-LSTM model shows a similar pattern to the standard LSTM, with limited generalisation ability and only moderate forecasting performance. Although its overall accuracy metrics are worse than those of the LSTM model, the CNN-LSTM appears to capture the underlying trend more effectively in some cases, which is likely due to the additional feature extraction performed by the CNN component.

Table 11: CNN-LSTM Model Loss Criteria Results

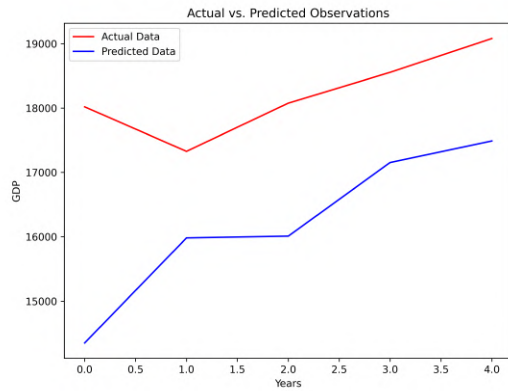
Country	MAE	MAPE	MSE
United States	12876.1	18.1%	166620540.1
Russian Federation	14771.4	38.6%	223320710.1
China	18033.9	87.4%	327593500.0
Brazil	2013.6	11.1%	4802349.1
Overall testing dataset results:	1.0792		2.3118



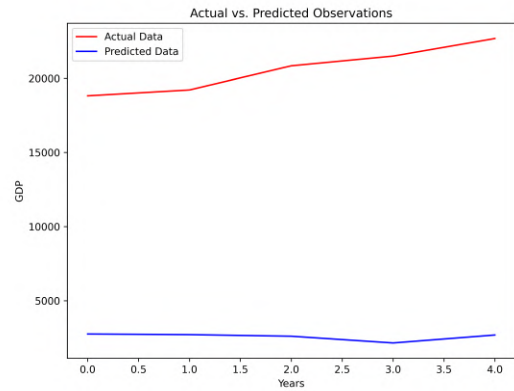
(a) United States - Actual vs. Predicted CNN-LSTM Results



(b) Russian Federation - Actual vs. Predicted CNN-LSTM Results



(c) Brazil - Actual vs. Predicted CNN-LSTM Results



(d) China - Actual vs. Predicted CNN-LSTM Results

Figure 7: Actual vs. Predicted CNN-LSTM Results

Table 12: Actual vs Predicted Values by Country for CNN-LSTM

Country	Type	Year 1	Year 2	Year 3	Year 4	Year 5
United States	Predicted	55413.2	56180.3	58422.2	59610.0	60593.7
	Actual	69112.6	67342.1	71307.4	72679.3	74158.7
Russian Federation	Predicted	24896.8	23668.0	23183.5	23633.6	21197.1
	Actual	37319.3	36376.3	38638.4	38214.5	39887.3
China	Predicted	2752.0	2709.5	2602.1	2147.4	2684.7
	Actual	18820.5	19215.2	20843.3	21499.4	22687.1
Brazil	Predicted	14351.6	15982.8	16010.3	17153.8	17489.2
	Actual	18018.6	17327.5	18075.7	18554.0	19079.8

Transformer

The model selection for Transformer was performed for 675 iterations over the following set of hyperparameters:

- Size of vector representation: [24, 32, 48, 64, 128]
- Number of attention heads: [2, 4, 8]
- hidden layer size of feedforward network: [24, 32, 48, 64, 128]
- Number of stacked transformers: [2, 4, 8]
- Number of epochs: [5, 10, 20]

Table 13: Top 5 GridSearch Results of the Transformer Model

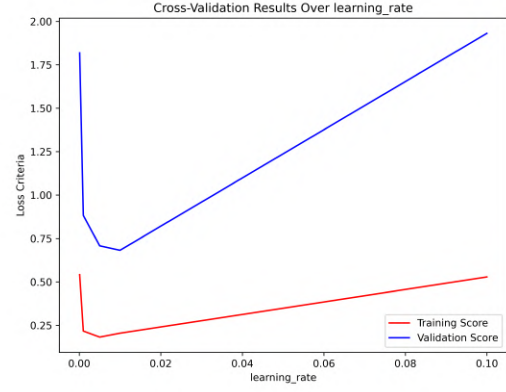
Features After Embed.	Attention Head	Size of Feedforw. Net.	Num. of Layers	Num. of Epochs	Train Score	Validation Score
32	2	24	2	5	0.2313	0.5842
24	4	24	2	20	0.1890	0.5961
24	4	48	2	20	0.1685	0.6318
32	2	24	2	10	0.1939	0.6323
32	8	32	2	10	0.2037	0.6333

The GridSearch identified two top-performing models based on validation performance. The selected configuration includes 32 features after embedding, 2 attention heads, a feed-forward size of 24, 2 encoder layers, and 5 epochs. However, based on Table 13 the training loss is substantially lower than the validation loss across the top results, indicating limited generalisation. Cross-validation was then performed across the number of epochs, batch size and learning rate to further tune the hyperparameters.

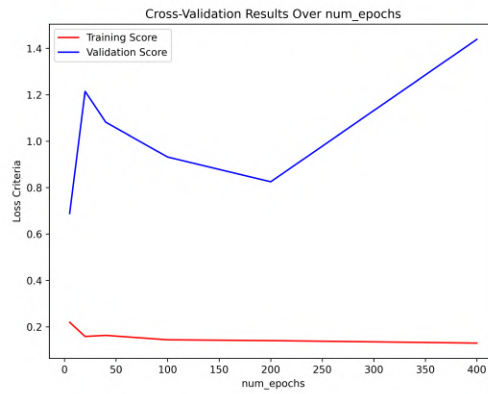
The results of cross-validation presented in Figure 8 display a decrease in loss with increasing batch size, reaching a minimum at moderate values before worsening beyond 100. The learning rate displays high sensitivity, with loss increasing beyond 0.01, making it the optimal choice. Increasing the number of epochs led to clear overfitting, as training loss decreased while validation loss diverged, indicating that 5 epochs provided the best balance. Furthermore, 200 epochs were tested on the validation data as the second best option, however it showed deteriorating results and was not used for the optimal model configuration.



(a) CV MSE Scores Across Batch Size



(b) CV MSE Scores Across Learning Rate



(c) CV MSE Scores Across A Number of Epochs

Figure 8: Transformer CV MSE Scores Across Hyperparameters

Based on the combination of GridSearch and cross-validation results, the parameters used for the final model configuration are the following:

- Size of vector representation: 24
- Number of attention heads: 4
- Hidden layer size of feed forward network: 24
- Drop out: 0.2
- Number of stacked encoders: 2
- Number of epochs: 5
- Learning rate: 0.01
- Batch size: 64

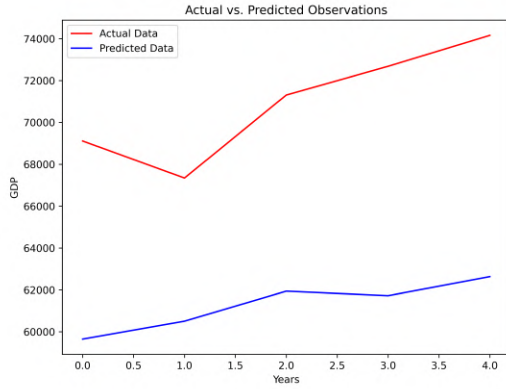
The final Transformer model achieved the best overall performance among the three models, with an MAE of 0.95. A comparison between the Transformer and the second-best model, the LSTM, across the reported countries showed the following results. For the USA, the Transformer demonstrated a similar ability to capture the overall trend as the CNN-LSTM, while providing a better estimate of the GDP level, resulting in the same MAPE as the LSTM. For the other three countries, a slight improvement of approximately 5% to 10% was observed in each

case. In particular, China achieved a MAPE of 2.7%, indicating excellent modelling of both the overall trend and short-term fluctuations. However, the forecasting performance for Russia and Brazil remained weak, although the Transformer showed a slightly better representation of the general trend. Overall, the obtained results suggest that the model is able to learn temporal dependencies between features through positional encoding, but it still fails to produce reliable forecasts in most cases.

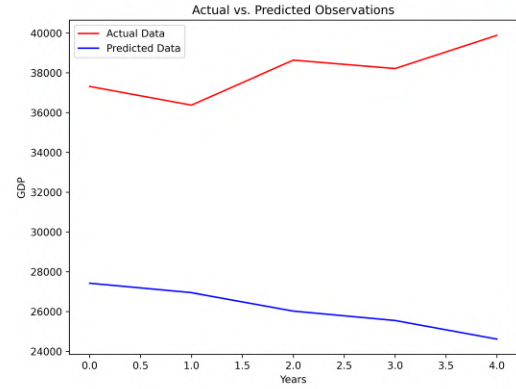
Overall, the comparison of the three deep learning models suggests that the Transformer provided the strongest forecasting performance, achieving the lowest overall MAE and more consistent improvements across most countries, while being the simplest model with only 5 epochs on training. The LSTM captured some temporal patterns but often failed to follow the overall GDP trend, limiting its generalisation. The CNN-LSTM did not improve the overall accuracy and generally performed slightly worse than LSTM, although it captured the shape of the trend better for some countries, particularly Brazil. The result indicated that the positional encoding analysed by the multiple self-attention heads in Transformer tends to be more effective in learning temporal dependencies within the sequential data. However, all three models still showed limited forecasting stability across countries, which may be due to the mixing countries in the same dataset, so further investigation such as adding country code as feature can improve the performance.

Table 14: Transformer Model Loss Criteria Results

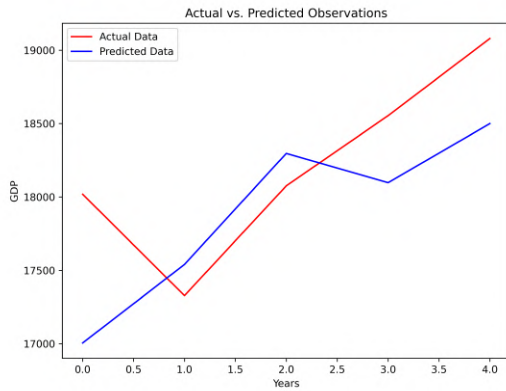
Country	MAE	MAPE	MSE
United States	9629.7	13.5%	95381115.4
Russian Federation	11978.5	31.3%	148001230.6
China	17745.2	86.0%	317566370.3
Brazil	496.8	2.7%	333318.8
Overall testing dataset results:	0.9507		2.0512



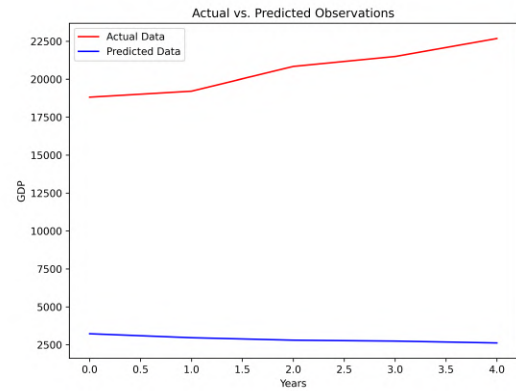
(a) United States - Actual vs. Predicted Transformer Results



(b) Russian Federation - Actual vs. Predicted Transformer Results



(c) Brazil - Actual vs. Predicted Transformer Results



(d) China - Actual vs. Predicted Transformer Results

Figure 9: Actual vs. Predicted Transformer Results

Table 15: Actual vs Predicted Values by Country for Transformer

Country	Type	Year 1	Year 2	Year 3	Year 4	Year 5
United States	Predicted	59648.9	60505.5	61941.0	61722.0	62634.2
	Actual	69112.6	67342.1	71307.4	72679.3	74158.7
Russian Federation	Predicted	27419.7	26946.3	26021.2	25546.0	24610.1
	Actual	37319.3	36376.3	38638.4	38214.5	39887.3
China	Predicted	3224.0	2961.8	2798.4	2739.0	2616.3
	Actual	18820.5	19215.2	20843.3	21499.4	22687.1
Brazil	Predicted	17004.9	17540.1	18296.6	18097.1	18499.9
	Actual	18018.6	17327.5	18075.7	18554.0	19079.8

Question 4 - Variational Autoencoder VAE

The VAE model architecture is presented by the encoder and decoder modules. The encoder is used to learn the parameters on latent distribution, to further generate latent space from it. This latent space is considered as the compressed representation of the original data. The Decoder part is used to transform sampled latent points to original scale, which is considered as sampling from the original data. The following question is a preparation to further use VAE as resampling technique to augment the existing data. The VAE model training is done by loss minimization as in all previously considered model. The loss for this model is presented by the summation of reconstruction loss and KL divergence multiplied by beta. In case of data sampling, KL divergence is a very important metric, thus the further model selection process is not only minimizing the overall loss metric but also the divergence term. The model selection for the Variational Autoencoder model was performed for over 800 iterations of grid search over the following set of parameters:

- Encoder specifications: [1, 2, 3, 4, 5] (shown in Table 16)
- Decoder specifications: [1, 2, 3, 4, 5] (shown in Table 16)
- Latent dimensions: [8, 16, 32, 64]
- Number of epochs: [5, 10, 40, 100]
- Learning rate: [0.0001, 0.01]

Table 16: Encoder/Decoder Specifications

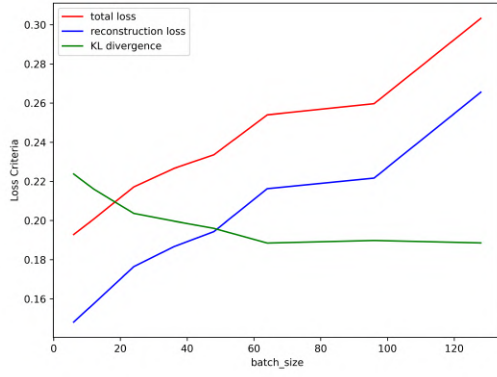
Specification	Number of Layers	Encoder/Decoder Layers
1	2	[{'linear_output':64, 'is_relu':True}, {'linear_output':32, 'is_relu':False}]
2	2	[{'linear_output':32, 'is_relu':True}, {'linear_output':16, 'is_relu':False}]
3	1	[{'linear_output':64, 'is_relu':True}]
4	1	[{'linear_output':64, 'is_relu':True}, {'linear_output':32, 'is_relu':True}]
5	2	[{'linear_output':128, 'is_relu':True}, {'linear_output':64, 'is_relu':False}]

Cross-validation was applied to grid search results to find the optimal model across batch size, learning rate and the number of epochs, shown in Figure 10, to further tune the hyperparameters.

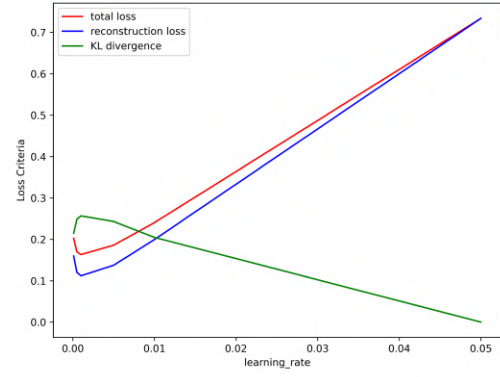
Table 17: Top 5 GridSearch Results of the VAE Model

Encoder Specs	Decoder Specs	Latent Dim	Epochs	Learning Rate	Total Loss	Recon Loss	KL Loss
1	1	64	100	0.0001	0.2114	0.1697	0.2085
2	1	64	100	0.0001	0.2149	0.1730	0.2097
1	2	64	100	0.0001	0.2210	0.1820	0.1949
2	2	64	100	0.0001	0.2287	0.1904	0.1916
2	1	64	100	0.0001	0.2555	0.2208	0.1738

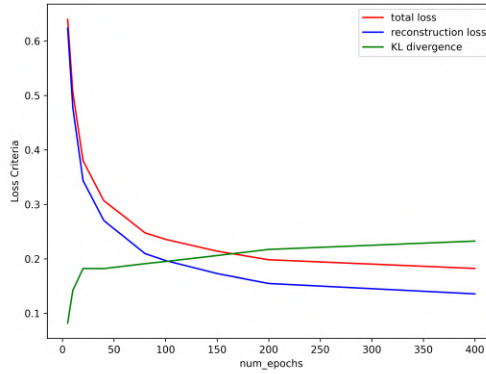
The cross-validation process has identified the clear trade-off between the reconstruction loss and KL divergence. This means that target Normal distribution is too simple to explain complex data and provide reliable reconstruction. This observation leads to the conclusion that optimal batch size, learning rate and epochs have to be selected based on achieving a balance between accurate reconstruction and appropriate latent-space regularisation.



(a) CV MSE Scores Across Batch Size



(b) CV MSE Scores Across Learning Rate



(c) CV MSE Scores Across A Number of Epochs

Figure 10: Autoencoder CV MSE Scores Across Hyperparameters

Therefore, the optimal configuration of the VAE model, where the initial 115 features were reduced to 64 latent dimensions, is as follows:

- Encoder specification: 1
- Decoder specification: 1
- Latent dimensions: 64
- Number of epochs: 200
- Learning rate: 0.005

Figure 11 illustrates the training process of the selected VAE model across total loss, reconstruction loss, and KL divergence. The results show a rapid decrease in total and reconstruction loss during the initial training stages, with a gradual stabilisation thereafter. In contrast, the KL divergence increases early in training and then stabilises, indicating that the latent space becomes gradually regularised. Furthermore, the plot confirms the choice of the batch size, since the losses seem to stabilise as they reach 200. The obtained results are confirming the observations made on the cross-validation stage, revealing that achieving ideal normal latent space is not best practice for for good reconstruction.

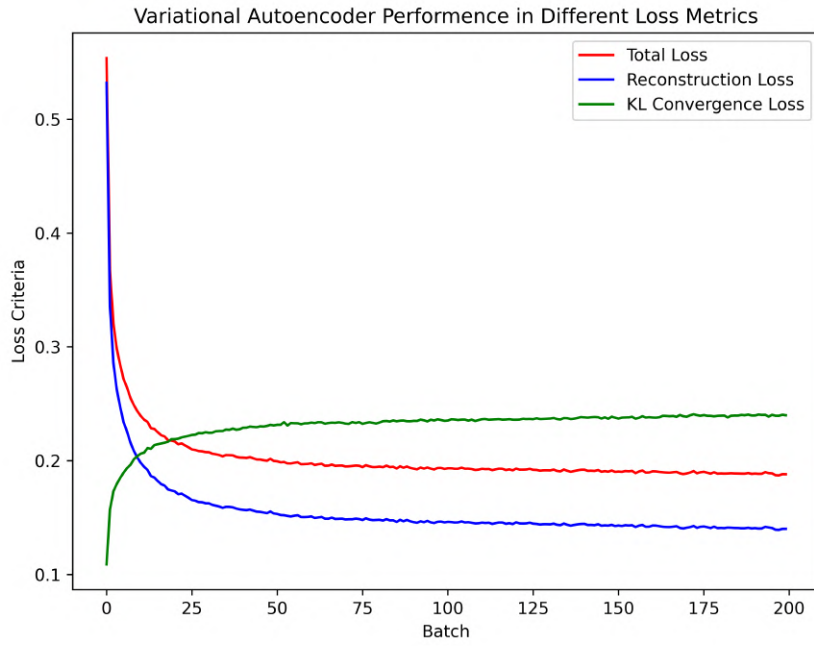


Figure 11: Training Process for the Selected Autoencoder Model for Reconstruction, KL Convergence and Total Losses

The Figure 12 is presenting the latent space reduced to 3D dimensionality. The results are clearly showing concentration around zero, with no clear clusters or outliers. Overall it can be stated that model have been able to capture distribution of the data.

3D Latent Representation

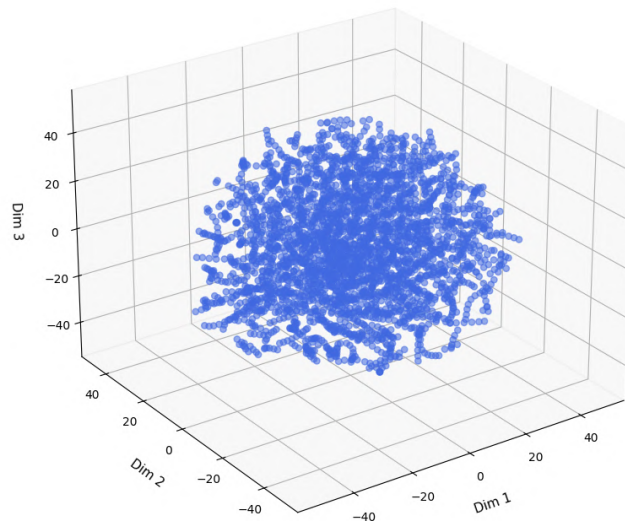


Figure 12: 3D Latent Representation

Question 5 - Comparison of Forecasts with and without Data Augmentation

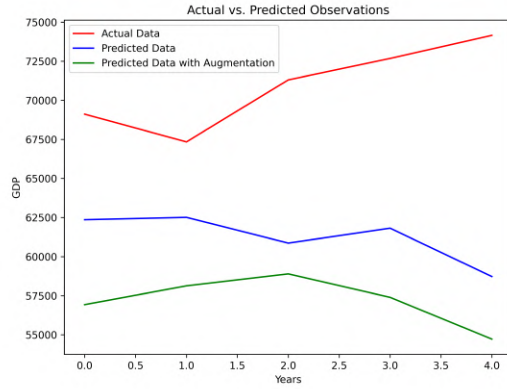
The VAE model selected in the previous question was used to simulate data from the latent space and bring it to original scale by decoder. The selection process included simulating 500, 1000, 2000, 4000 and 10000 of additional rows with further evaluation on a validation set. Overall the difference in amount of augmented data tend to show trade-off in performance between transformer and other 2 models. Based on the obtained results from question 4, we have added 2000 of additional rows of data, which is approximately 50% of the original training dataset, as an approach that lead to best results.

LSTM Augmented versus original

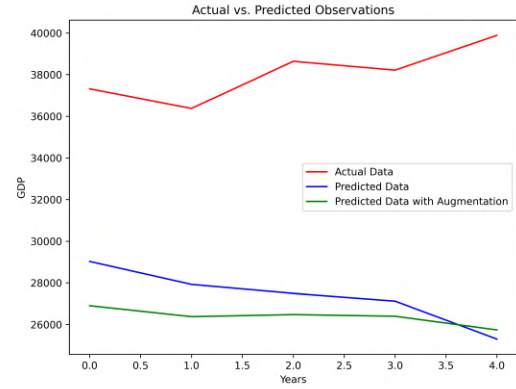
The inclusion of 2,000 additional observations in the training set led to an improvement in performance across the full testing dataset. The notable reduction in MSE indicates a decrease in the magnitude of large forecasting errors. However, the model's performance deteriorated for the USA and Russia, while improvements were observed for China and Brazil, suggesting that the use of the Variational Autoencoder produced mixed and somewhat suboptimal results, with improvements in overall performance.

Table 18: LSTM Model Loss Criteria Results for Original and Augmented Data

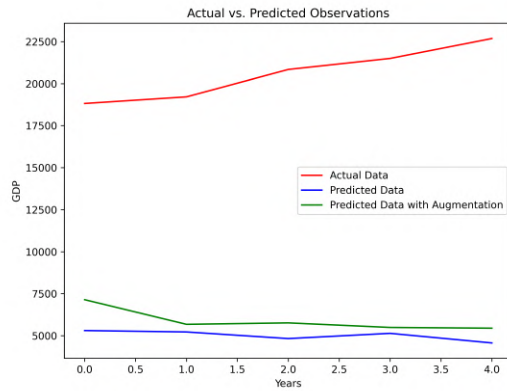
Country	Original Data			Augmented Data		
	MAE	MAPE	MSE	MAE	MAPE	MSE
United States	9664.3	13.5%	106835587.9	13707.0	19.2%	199788023.1
Russian Federation	10712.2	28.0%	120021113.2	11707.6	30.7%	139220964.8
China	15598.1	75.5%	246108418.1	14705.9	71.0%	220025182.9
Brazil	2471.4	13.2%	11692866.6	809.3	4.4%	772642.1
Overall testing dataset results:	1.05		2.2665	0.94		1.8347



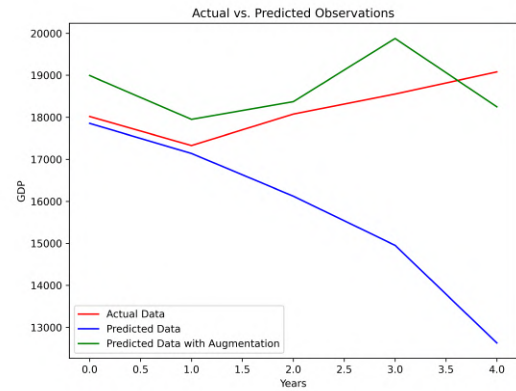
(a) US Forecasting Results Original vs Augmented



(b) Russia Forecasting Results Original vs Augmented



(c) China Forecasting Results Original vs Augmented

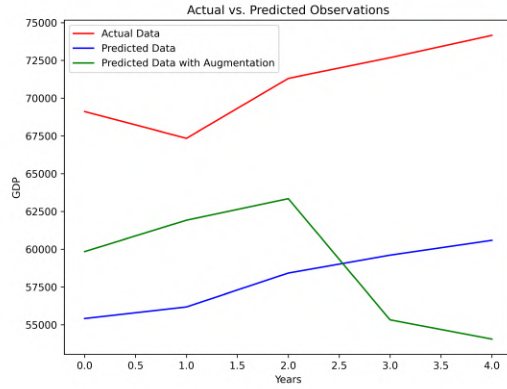


(d) Brazil Forecasting Results Original vs Augmented

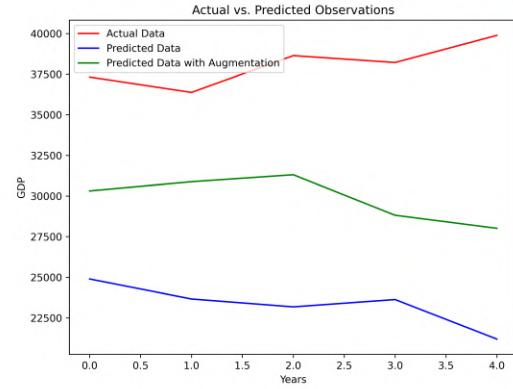
Figure 13: Forecasting Results Original vs Augmented

CNN-LSTM Augmented versus original

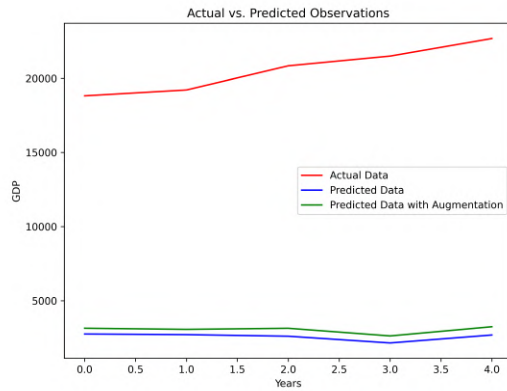
The CNN-LSTM model showed some improvement in overall forecasting performance. The MAE for the testing dataset decreased from 1.08 to 1.00, indicating that the augmented data had a positive effect on the model's predictive ability. The forecasts for the USA and China improved slightly, with better capture of the underlying trend, whereas the performance for Brazil deteriorated by 0.3%. The most notable change was observed for Russia, where the MAPE changed by 17%, accompanied by a clear visual improvement in both trend and level estimation.



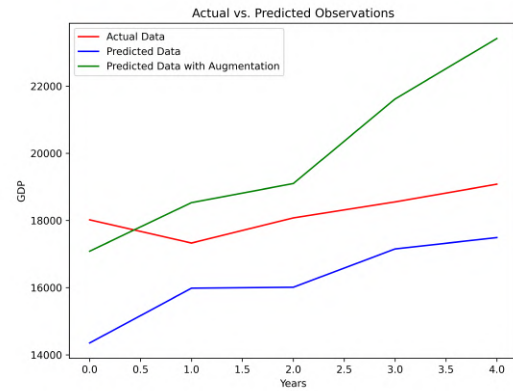
(a) US Forecasting Results Original vs Augmented



(b) Russia Forecasting Results Original vs Augmented



(c) China Forecasting Results Original vs Augmented



(d) Brazil Forecasting Results Original vs Augmented

Figure 14: Forecasting Results Original vs Augmented

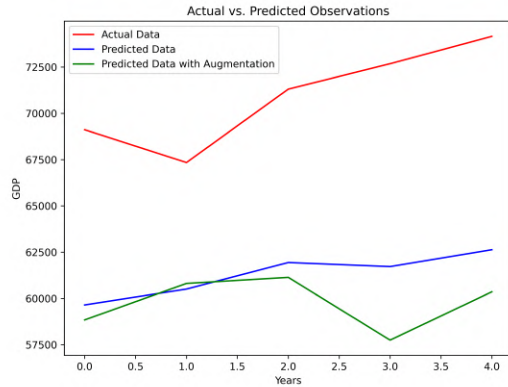
Table 19: CNN-LSTM Model Loss Criteria Results for Original and Augmented Data

Country	Original Data			Augmented Data		
	MAE	MAPE	MSE	MAE	MAPE	MSE
United States	12876.1	18.1%	166620540.1	12022.0	16.7%	176831587.5
Russian Federation	14771.4	38.6%	223320710.1	8220.0	21.4%	72457965.8
China	18033.9	87.4%	327593500.0	17570.4	85.2%	310886508.0
Brazil	2013.6	11.1%	4802349.1	2112.9	11.4%	6318048.3
Overall testing dataset results:	1.0792		2.3118	1.0021		2.2269

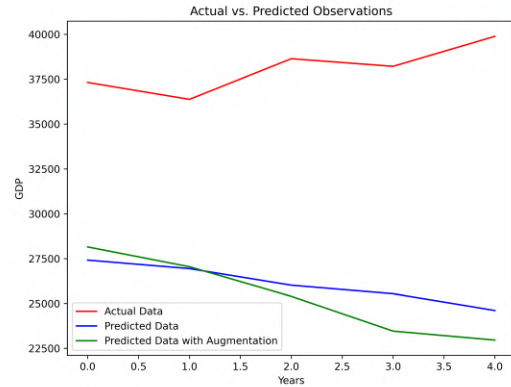
Transformer Augmented versus original

The performance of the Transformer model deteriorated after the inclusion of augmented data. The MAE for the testing dataset increased from 0.95 to 1.01, indicating that the augmented data had a negative impact on the overall forecasting performance. At the country level, the USA and Russian Federation both experienced a decline in performance, with increases in MAPE and

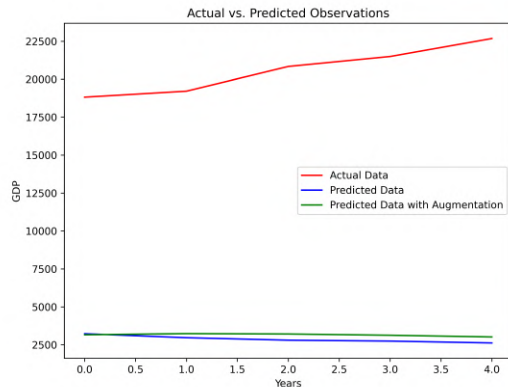
poorer alignment with the true GDP levels. China showed a slight improvement, with MAPE decreasing from 86.0% to 84.6%, although the model still struggled to accurately capture the overall scale of the data. Brazil experienced the most significant deterioration, with MAPE increasing from 2.7% to 6.6%, alongside a clear overestimation of both trend and level. Overall, these results suggest that the augmented data did not enhance the Transformer’s ability to generalise and, in most cases, led to worse forecasting performance.



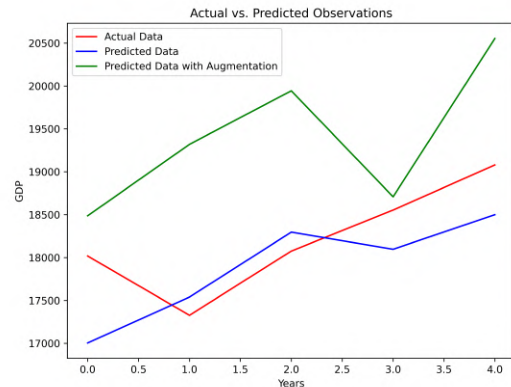
(a) US Forecasting Results Original vs Augmented



(b) Russia Forecasting Results Original vs Augmented



(c) China Forecasting Results Original vs Augmented



(d) Brazil Forecasting Results Original vs Augmented

Figure 15: Forecasting Results Original vs Augmented

Table 20: Transformer Model Loss Criteria Results for Original and Augmented Data

Country	Original Data			Augmented Data		
	MAE	MAPE	MSE	MAE	MAPE	MSE
United States	9629.7	13.5%	95381115.4	11140.7	15.6%	132991912.3
Russian Federation	11978.5	31.3%	148001230.6	12682.7	33.1%	170102485.1
China	17745.2	86.0%	317566370.3	17467.0	84.6%	307330769.7
Brazil	496.8	2.7%	333318.8	1191.5	6.6%	1974835.5
Overall testing dataset results:	0.9507		2.0512	1.0052		2.3815

Overall, the inclusion of data augmentation produced mixed results across the three forecasting models. For the LSTM and CNN-LSTM, augmentation improved performance on the overall testing dataset, indicating that the additional synthetic observations helped reduce large prediction errors and improved the models ability to learn useful patterns from the data. However, these improvements were not consistent across all countries, as some series still showed deteriorated performance. In contrast, the Transformer model performed worse after augmentation, suggesting that the synthetic data did not support this framework in the same way. Multiple tests has shown that with increased percent of augmented data this model tend to provide better performance.

Therefore, the results indicate that data augmentation can be beneficial in this setting, but its effectiveness depends on the model architecture and is not sufficiently stable to guarantee improvements across all forecasting tasks. Feature extraction technique in CNN-LSTM and embedding in Transformer are tend to show better results with augmented data but may fail to generalize to the testing data as it happens for Transformer.

Results

Multiple tasks, including classification, forecasting, and data augmentation, were considered within the scope of this report. In addition to the main augmentation framework, an additional experiment based on simple forward and backward fill was also implemented. This approach led to a deterioration in MLP classification performance, although it produced slight forecasting improvements for some individual countries across all models. Among the three forecasting models considered, the Transformer was the most computationally efficient while also delivering the strongest overall forecasting performance on the original data. However, the inclusion of Variational Autoencoder-based augmentation improved the overall results of the LSTM and CNN-LSTM models, while leading to a deterioration in the Transformer's performance. The tests performed on the amount of augmented data, resulted in ability of each model to improve from data augmentation with careful validation process. Biggest improvements were observed to Transformer and CNN-LSTM, most likely due to the feature extraction. Overall, the results show that no single approach was optimal across all tasks, and that the usefulness of augmentation depends not only on the quality of generated data but also on the learning behaviour of the selected model architecture.

Appendix A

USA plots for every indicator in the final dataset:

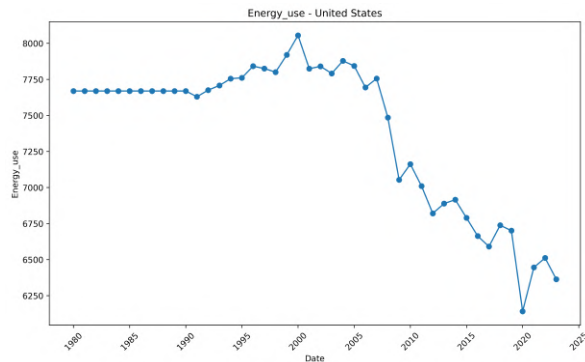


Figure A.1: USA Energy Use

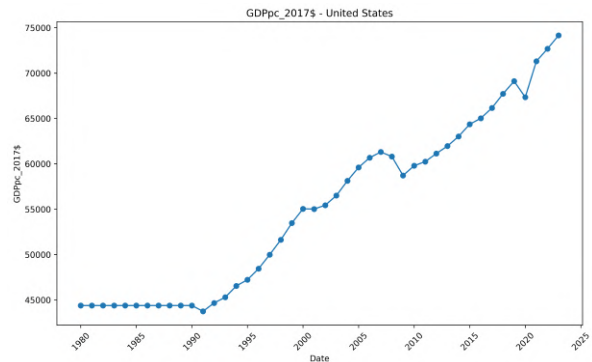


Figure A.2: USA GDP

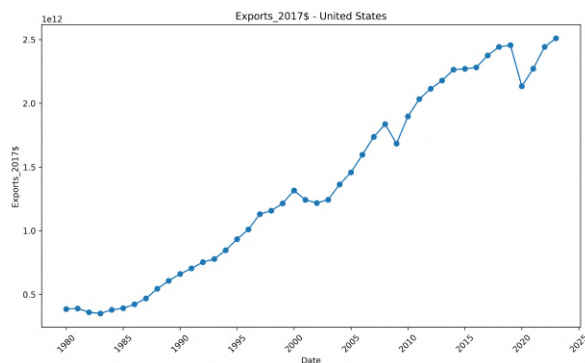


Figure A.3: USA Exports

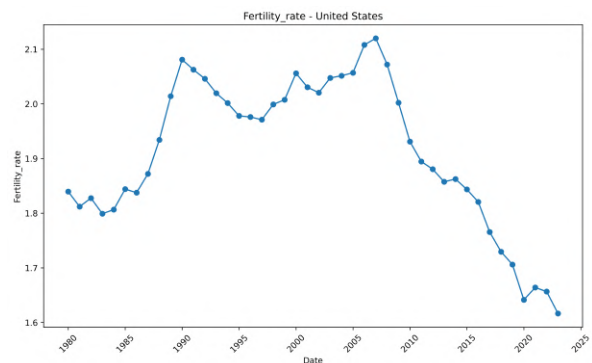


Figure A.4: USA Fertility Rate

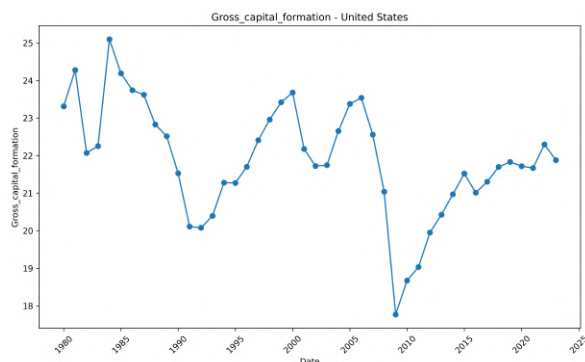


Figure A.5: USA Gross Capital Formation

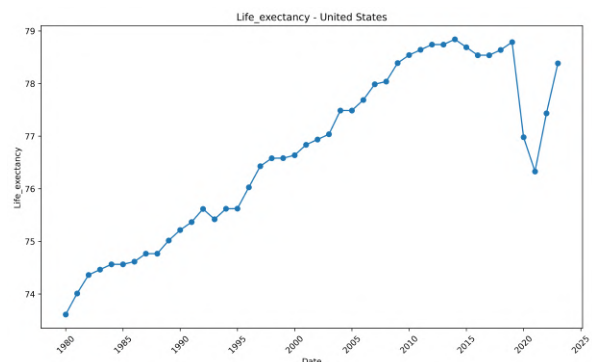


Figure A.6: USA Life Expectancy

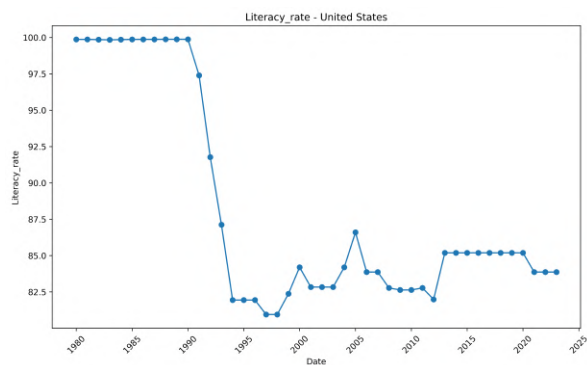


Figure A.7: USA Literacy Rate

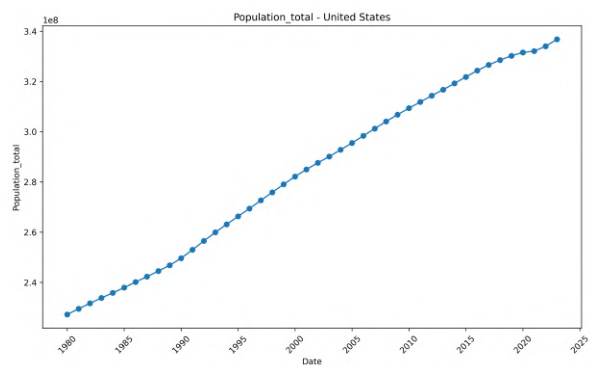


Figure A.8: USA Population Total

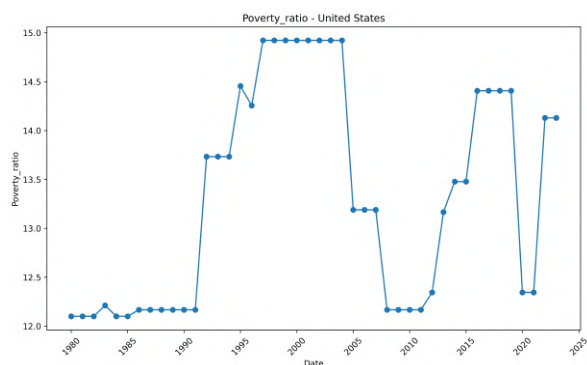


Figure A.9: USA Poverty Ratio

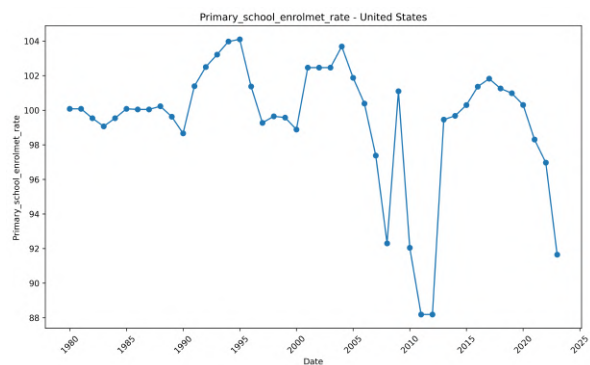


Figure A.10: USA Primary School Enrolment Rate

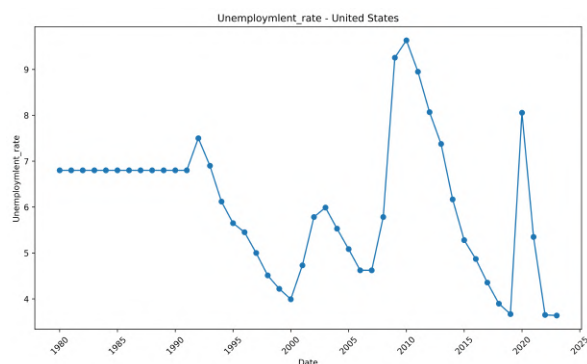


Figure A.11: USA Unemployment Rate

China plots for every indicator in the final dataset:

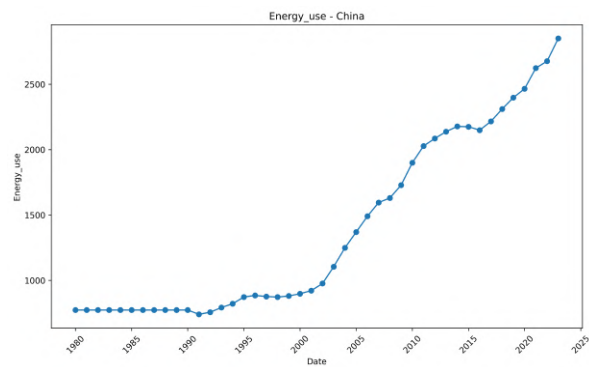


Figure A.12: China Energy Use

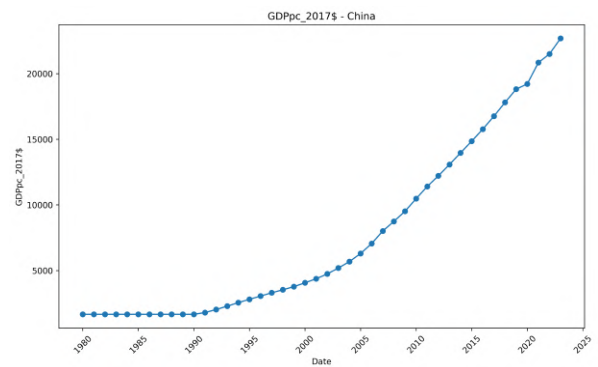


Figure A.13: China GDP

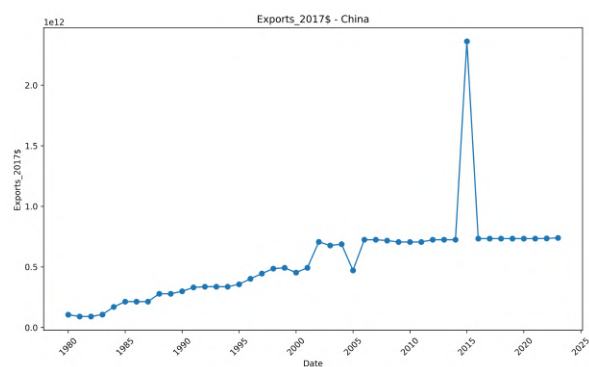


Figure A.14: China Exports

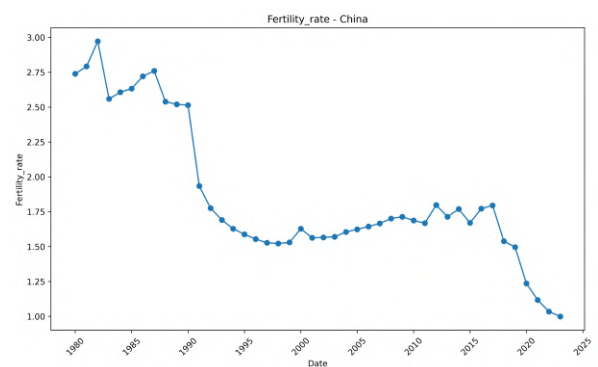


Figure A.15: China Fertility Rate

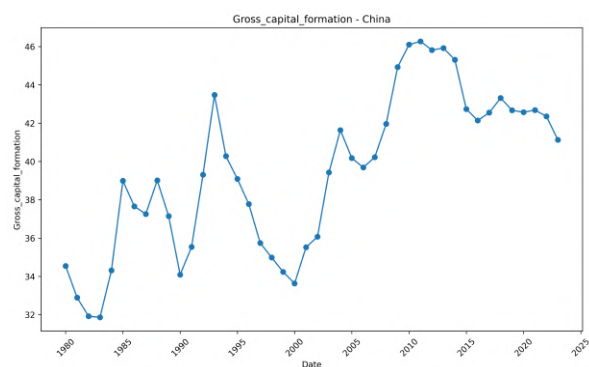


Figure A.16: China Gross Capital Formation

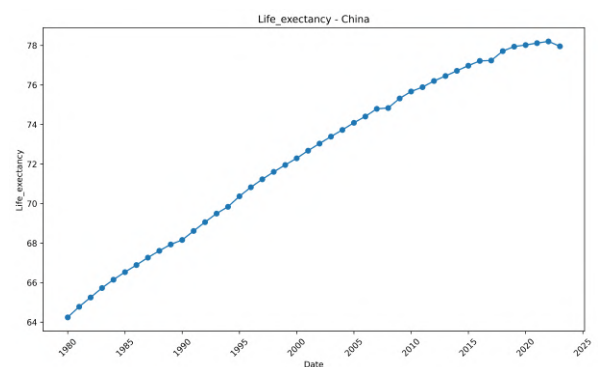


Figure A.17: China Life Expectancy

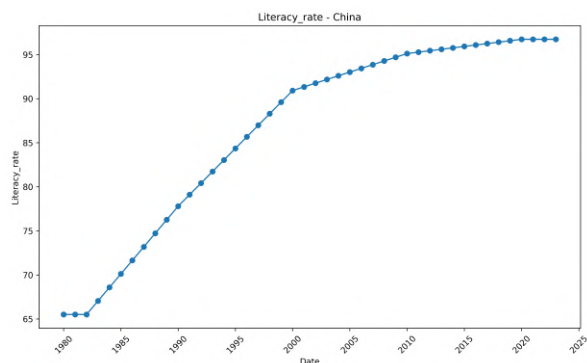


Figure A.18: China Literacy Rate

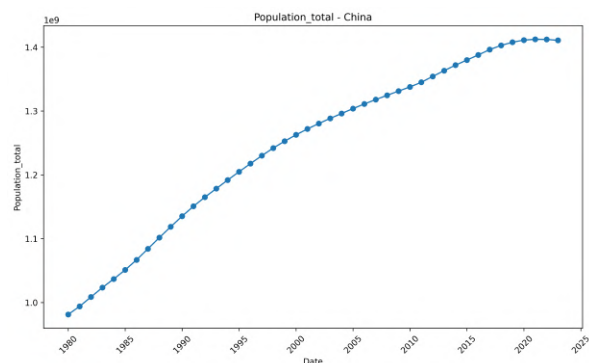


Figure A.19: China Population Total

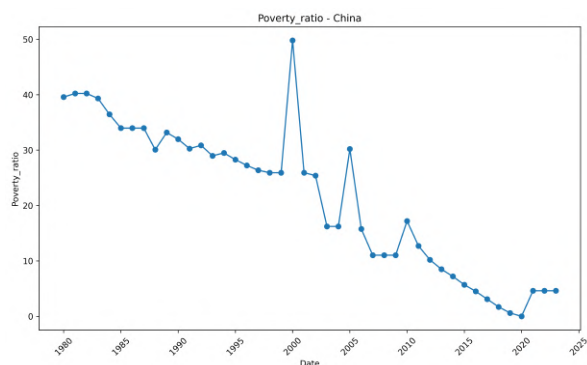


Figure A.20: China Poverty Ratio

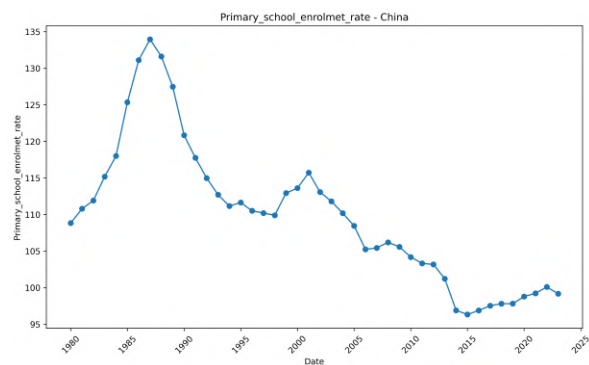


Figure A.21: China Primary School Enrolmet Rate

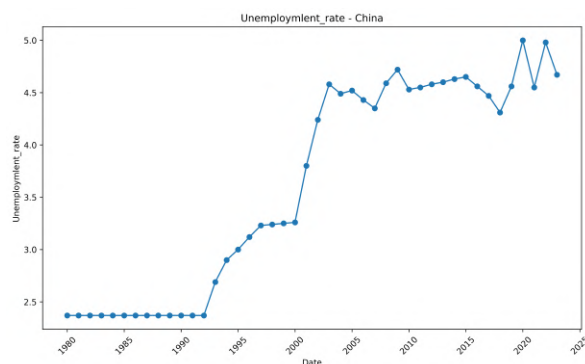


Figure A.22: China Unemployment Rate

Russia plots for every indicator in the final dataset:

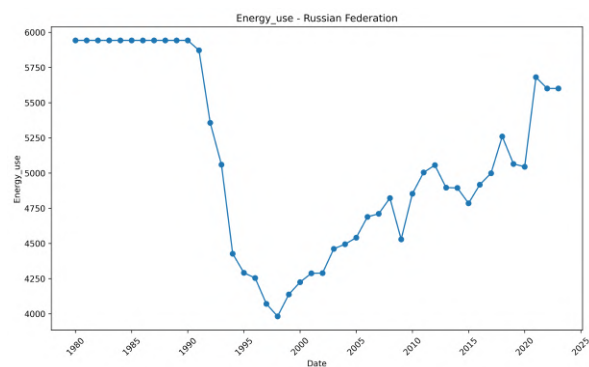


Figure A.23: Russian Federation Energy Use

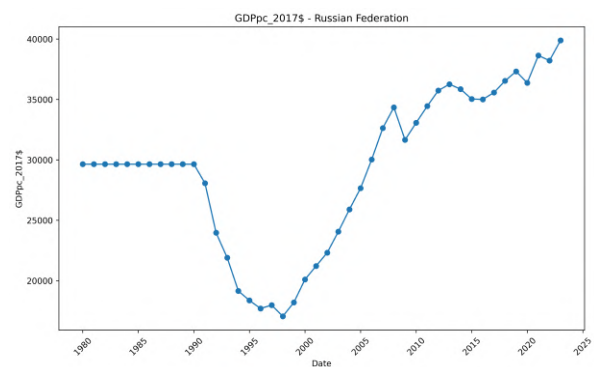


Figure A.24: Russian Federation GDP

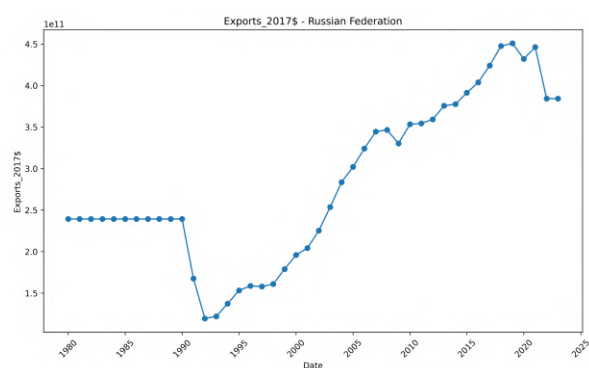


Figure A.25: Russian Federation Exports

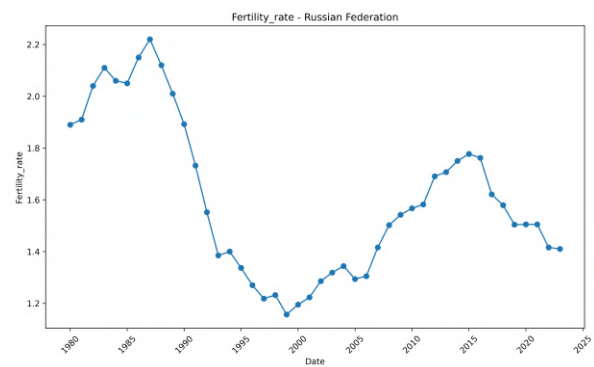


Figure A.26: Russian Federation Fertility Rate



Figure A.27: Russian Federation Gross Capital Formation

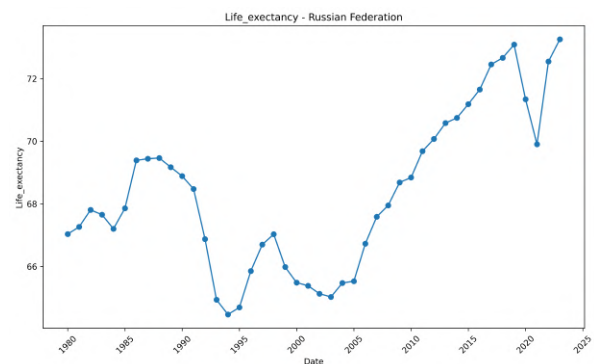


Figure A.28: Russian Federation Life Expectancy

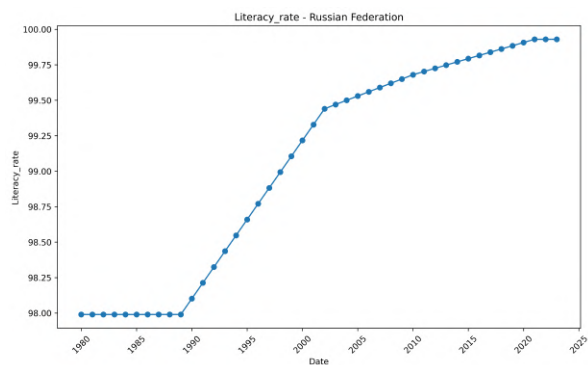


Figure A.29: Russian Federation Literacy Rate

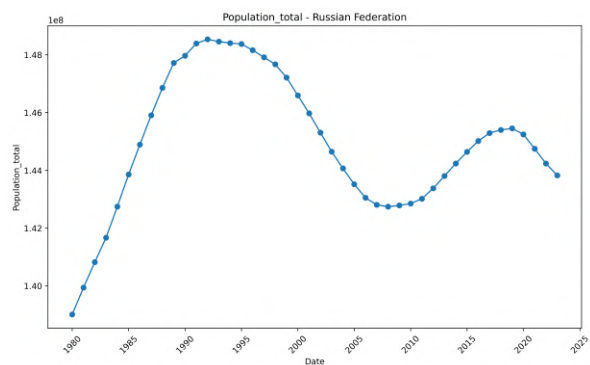


Figure A.30: Russian Federation Population Total

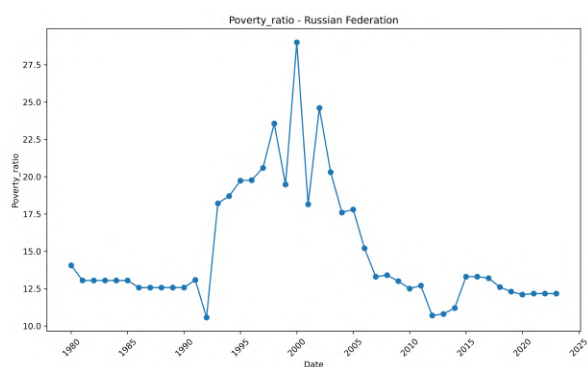


Figure A.31: Russian Federation Poverty Ratio

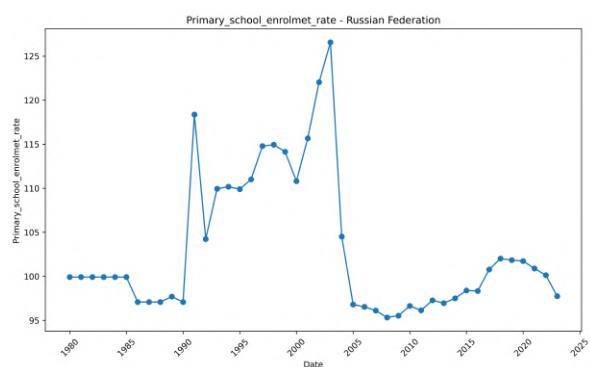


Figure A.32: Russian Federation Primary School Enrolment Rate

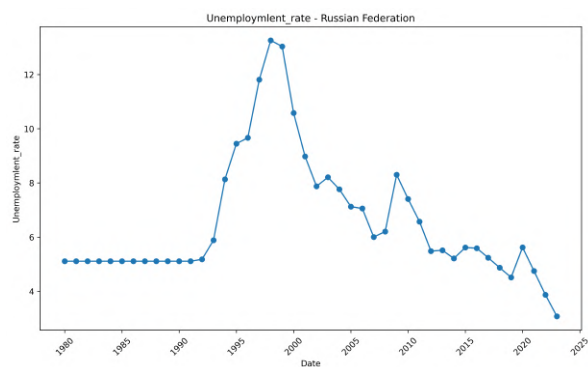


Figure A.33: Russian Federation Unemployment Rate

Brazil plots for every indicator in the final dataset:

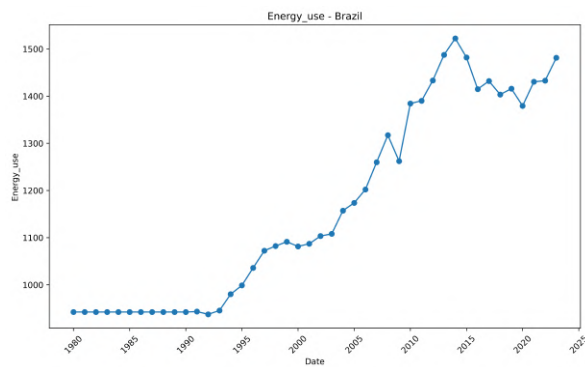


Figure A.34: Brazil Energy Use

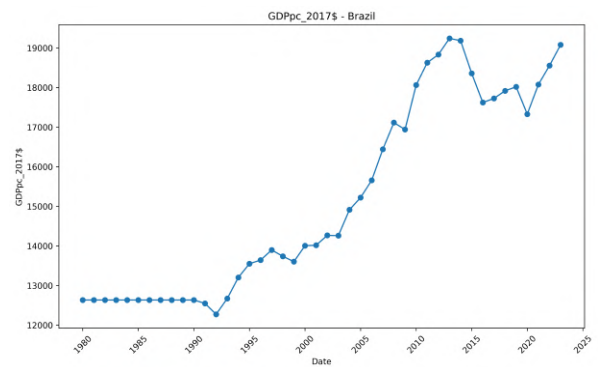


Figure A.35: Brazil GDP

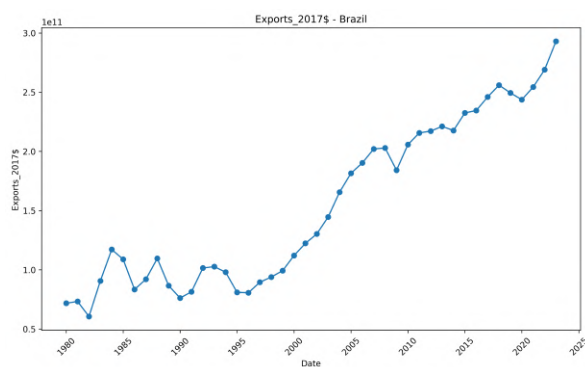


Figure A.36: Brazil Exports

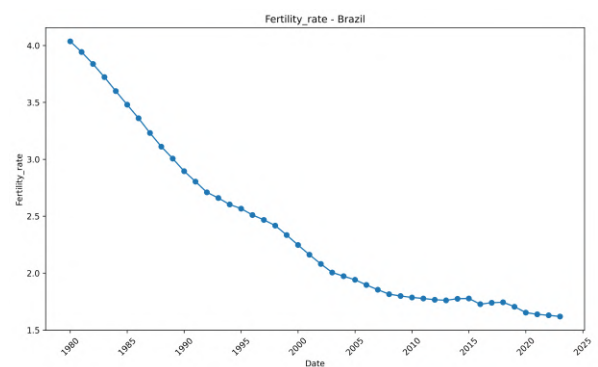


Figure A.37: Brazil Fertility Rate

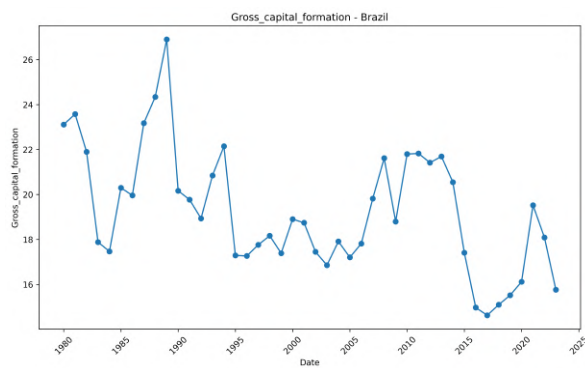


Figure A.38: Brazil Gross Capital Formation

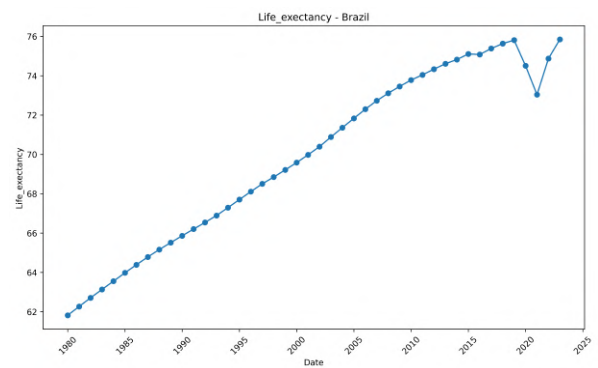


Figure A.39: Brazil Life Expectancy

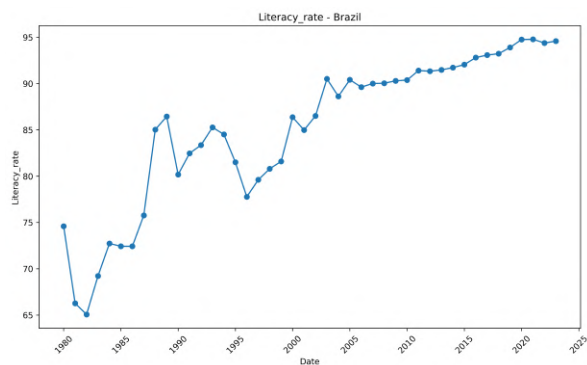


Figure A.40: Brazil Literacy Rate

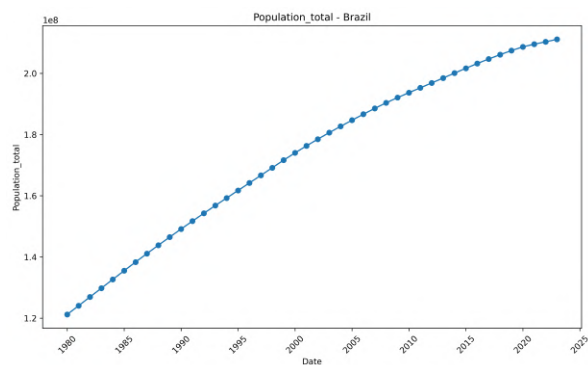


Figure A.41: Brazil Population Total

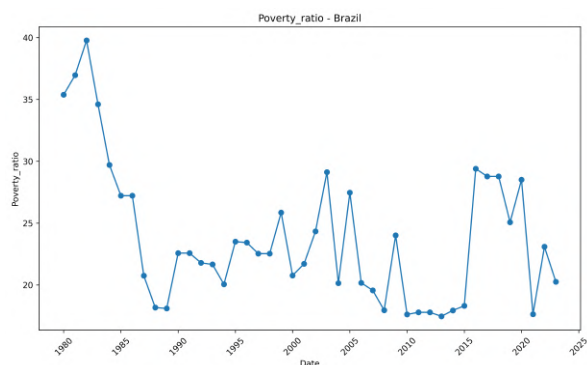


Figure A.42: Brazil Poverty Ratio

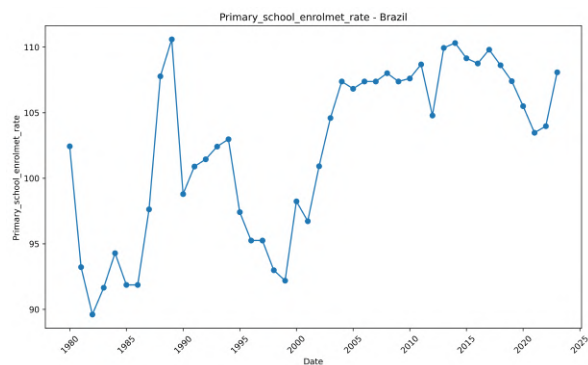


Figure A.43: Brazil Primary School Enrolment Rate

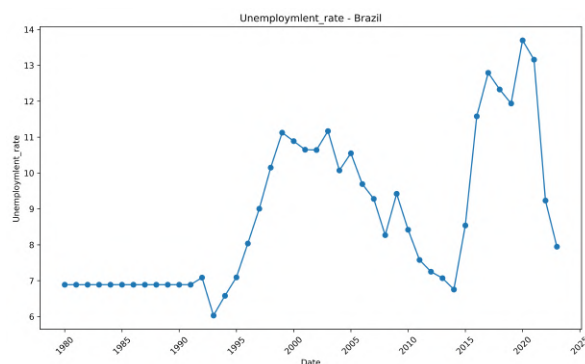


Figure A.44: Brazil Unemployment Rate